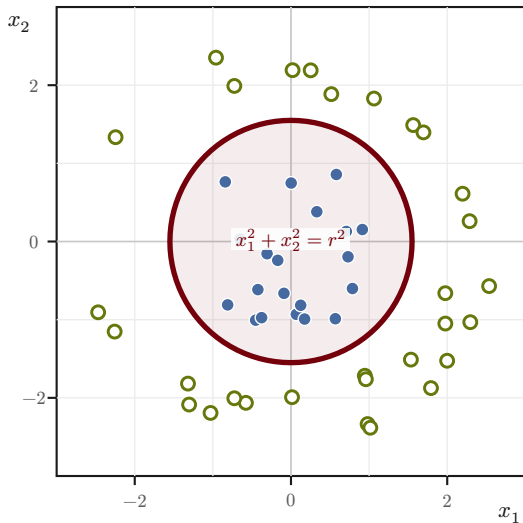


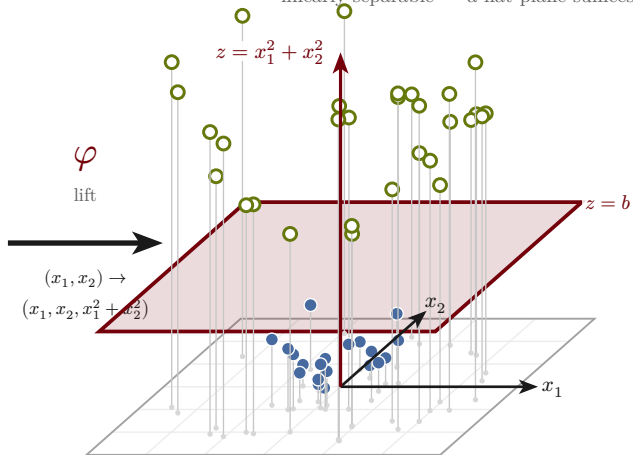
input space $\mathbb{R}^2$

not linearly separable — needs a curved boundary



feature space $\mathbb{R}^3$

linearly separable — a flat plane suffices



● class $\oplus$ (inner, low z)

○ class $\ominus$ (outer, high z)

— separator: curve $\leftrightarrow$ plane

kernel trick: a linear learner needs only $\langle \varphi(\mathbf{x}), \varphi(\mathbf{x}') \rangle = k(\mathbf{x}, \mathbf{x}')$ — so φ is never formed explicitly.